

1.

```
(mlops-week1) kai9735@fisfin:~$ cd bitamin-mlops-1-snapshot
(mlops-week1) kai9735@fisfin:~/bitamin-mlops-1-snapshot$ pip install -r requirements.txt
```

2.

```
(mlops-week1) kai9735@fisfin:~/bitamin-mlops-1-snapshot$ pip list
Package           Version
-----
cloudpickle       3.1.2
joblib            1.6.0
numpy             2.2.6
packaging         26.3
pandas            2.3.3
pip               26.2.1
python-dateutil   2.9.0.post0
pytz              2026.3.post1
scikit-learn      1.7.2
scipy             1.15.3
setuptools        83.0.0
six               1.17.0
threadpoolctl     3.6.0
tzdata            2026.3
wheel             0.47.0
```

3.

```
(mlops-week1) kai9735@fisfin:~/bitamin-mlops-1-snapshot$ python app.py
/home/kai9735/miniconda3/envs/mlops-week1/lib/python3.10/site-packages/sklearn/linear_model/_logistic.py:473: ConvergenceWarning: lbfgs failed to converge after 1000 iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

Increase the number of iterations to improve the convergence (max_iter=1000).
You might also want to scale the data as shown in:
    https://scikit-learn.org/stable/modules/preprocessing.html
Please also refer to the documentation for alternative solver options:
    https://scikit-learn.org/stable/modules/linear_model.html#logistic-regression
n_iter_i = _check_optimize_result(
Accuracy: 0.7854
```

4.

```
(mlops-week1) kai9735@fisfin:~/bitamin-mlops-1-snapshot$ docker images
```

IMAGE	ID	DISK USAGE	Info → CONTENT SIZE	U In Use EXTRA
bitamin-mlops-1:latest	b756ad1f70f8	642MB	147MB	

5.

```
(mlops-week1) kai9735@fisfin:~/bitamin-mlops-1-snapshot$ docker run bitamin-mlops-1
/usr/local/lib/python3.10/site-packages/sklearn/linear_model/_logistic.py:473: ConvergenceWarning: lbfgs failed to converge after 1000 iteration(s) (status=1):
STOP: TOTAL NO. OF ITERATIONS REACHED LIMIT

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